

Order Delivery Time Prediction

Delivery Time Estimation Project Summary

Objective

To build a regression model that accurately predicts order delivery time based on various order-related features. The model aims to support logistics teams in enhancing delivery efficiency, reducing customer wait times, and improving operational planning.

Dataset Overview

- market_id: Identifier for the market/zone where the order originates
- created_at: Timestamp of order placement
- actual_delivery_time: Timestamp when the order was delivered
- store_primary_category: Type of restaurant (e.g., Fast Food, Casual Dining)
- order_protocol: Method used to place the order (e.g., via app, phone call)
- total_items: Number of items in the order
- subtotal: Price of the order (before taxes, tips, etc.)

Exploratory Data Analysis (EDA)

- Delivery Time Distribution: Skewed to the right (some unusually long deliveries)
- Total Items vs. Delivery Time: Positive correlation - larger orders tend to take longer
- Subtotal Correlation: Higher subtotal generally indicates longer prep time and delivery delay
- Order Protocol: Orders placed via app had faster and more consistent delivery times
- Market-level Variations: Some markets consistently experienced longer delivery delays

Modeling Approach

Model Used: Linear Regression

Target Variable: Delivery duration (difference between created_at and actual_delivery_time)

Features Used:

- total_items
- subtotal
- order_protocol
- Encoded store_primary_category

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- Encoded market_id

Performance Metrics

- R2 Score: 0.72 - 72% of variance explained by the model
- RMSE: 5.3 minutes - Typical prediction error is ± 5.3 minutes
- MAE: 4.1 minutes - Average absolute error across predictions

Key Insights

- Orders with more items and higher subtotals are more prone to delays
- App-based orders were delivered faster and more predictably
- Market location strongly affects delivery time
- Restaurant type influences preparation and wait times

Recommendations

1. Prioritize App-Based Orders: Encourage app-based ordering
2. Flag Complex Orders: Alert ops teams about possible delays
3. Geo-Aware Dispatching: Adjust based on market trends
4. Feature Expansion: Add weather, traffic, time-of-day data

Conclusion

This linear regression model provides a solid, interpretable baseline for predicting delivery time. With more data and refinement, it can help improve logistics, reduce delays, and enhance customer satisfaction.